

Operating Systems - Day 1 Revision Sheet

1. Program vs Process

- Keywords: Passive, Active, Execution, Memory, PCB, Resources
- Definition: A program is a passive set of instructions stored on disk, whereas a process is a program currently being executed with its own memory, CPU state, and resources.
- Follow-ups: Can multiple processes run the same program? Why does every process need a PCB?

2. Process Memory Layout

- Keywords: Text Segment, Data Segment, Heap, Stack, Dynamic Allocation, Function Calls
- Definition: A process's memory is divided into the text, data, heap, and stack segments.
- Follow-ups: Why does the stack grow downward? Why does the heap grow upward?

3. PCB

- Keywords: PID, Process State, Program Counter, CPU Registers, Scheduling Information, Memory Information, Context Switching
- Definition: PCB is a kernel data structure storing all information required to manage and resume a process.
- Follow-ups: Why is PCB stored in kernel memory? What happens if PCB is corrupted?

4. Process States

- Keywords: New, Ready, Running, Waiting, Terminated
- Definition: A process moves through lifecycle states depending on scheduling and events.
- Follow-ups: Why Running→Waiting? What causes Waiting→Ready?

5. Scheduler

- Keywords: Ready Queue, Scheduling Algorithm, CPU Allocation
- Definition: Selects the next process from the ready queue.
- Follow-ups: Types of schedulers?

6. Dispatcher

- Keywords: Context Switch, User Mode, Kernel Mode
- Definition: Gives CPU control to the selected process.
- Follow-ups: Scheduler vs Dispatcher?

7. Context Switching

- Keywords: Save Context, Restore Context, PCB, Registers, Program Counter
- Definition: Saves one process state and restores another.
- Follow-ups: Why is it overhead?

8. Timer Interrupt

- Keywords: Fairness, Time Slice, Preemption, Multitasking
- Definition: Ensures periodic scheduling decisions.
- Follow-ups: What if no timer interrupt?

9. Interrupt

- Keywords: Signal, CPU Attention, ISR
- Definition: A signal requesting immediate CPU attention.
- Follow-ups: Types of interrupts?

10. ISR

- Keywords: Interrupt Handler, Kernel Function, Event Handling
- Definition: Kernel routine executed to service an interrupt.
- Follow-ups: Why should ISR be short?

11. IVT / IDT

- Keywords: Interrupt Number, Handler Address, Lookup, Jump
- Definition: Maps interrupt numbers to ISR addresses.
- Follow-ups: IVT vs PCB?

12. Hardware vs Software Interrupts

- Hardware: Keyboard, Mouse, Disk, Timer
- Software: System Call, Trap, Kernel Service
- Definition: Hardware interrupts come from devices; software interrupts are generated by programs to request OS services.

13. User Mode vs Kernel Mode

- Keywords: Privileges, Protection, Hardware Access, System Calls
- Definition: User mode is restricted; kernel mode has full privileges.
- Follow-ups: Why can't user mode access hardware?

14. System Calls

- Keywords: Privileged Operations, File Access, Process Creation, Memory Allocation
- Definition: Controlled interface to request kernel services.
- Follow-ups: Examples?

15. Interrupt Priority

- Keywords: Priority Levels, Interrupt Controller, Nested Interrupts
- Definition: Higher-priority interrupts are serviced before lower-priority ones.
- Follow-ups: Who decides priority?

Revision Checklist

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- Process Memory Layout
- PCB
- Process States
- Scheduler
- Dispatcher
- Context Switching
- Timer Interrupt
- Interrupts
- ISR
- IVT/IDT
- Hardware vs Software Interrupts
- User Mode vs Kernel Mode
- System Calls
- Interrupt Priority